

# The algebraic monodromy of Lagrangian fibrations

Edward Varvak

Department of Mathematics, Science, and Computer Science  
University of Illinois Chicago

May 2026

# Holomorphic symplectic manifolds

## Definition

A *symplectic manifold*  $(M, \omega)$  is a smooth manifold  $M$  equipped with a closed non-degenerate global 2-form  $\omega \in H^0\left(M, \bigwedge^2 T^*M\right)$ .

Contracting with  $\omega$  induces an isomorphism  $\iota_\omega : TM \xrightarrow{\sim} T^*M$ .

# Holomorphic symplectic manifolds

## Definition

A *symplectic manifold*  $(M, \omega)$  is a smooth manifold  $M$  equipped with a closed non-degenerate global 2-form  $\omega \in H^0\left(M, \bigwedge^2 T^*M\right)$ .

Contracting with  $\omega$  induces an isomorphism  $\iota_\omega : TM \xrightarrow{\sim} T^*M$ .

## Examples

- $\mathbb{R}^{2n}$  admits a natural symplectic form  $\omega = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$ .

# Holomorphic symplectic manifolds

## Definition

A *symplectic manifold*  $(M, \omega)$  is a smooth manifold  $M$  equipped with a closed non-degenerate global 2-form  $\omega \in H^0\left(M, \bigwedge^2 T^*M\right)$ .

Contracting with  $\omega$  induces an isomorphism  $\iota_\omega : TM \xrightarrow{\sim} T^*M$ .

## Examples

- $\mathbb{R}^{2n}$  admits a natural symplectic form  $\omega = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$ .
- Any Kähler manifold  $X$  is symplectic with respect to the Kähler form.

# Holomorphic symplectic manifolds

## Definition

A *symplectic manifold*  $(M, \omega)$  is a smooth manifold  $M$  equipped with a closed non-degenerate global 2-form  $\omega \in H^0(M, \wedge^2 T^*M)$ .

Contracting with  $\omega$  induces an isomorphism  $\iota_\omega : TM \xrightarrow{\sim} T^*M$ .

## Examples

- $\mathbb{R}^{2n}$  admits a natural symplectic form  $\omega = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$ .
- Any Kähler manifold  $X$  is symplectic with respect to the Kähler form.
- Any projective manifold is symplectic with respect to the Fubini–Study metric.

# Holomorphic symplectic manifolds

## Definition

A *symplectic manifold*  $(M, \omega)$  is a smooth manifold  $M$  equipped with a closed non-degenerate global 2-form  $\omega \in H^0\left(M, \bigwedge^2 T^*M\right)$ .

Contracting with  $\omega$  induces an isomorphism  $\iota_\omega : TM \xrightarrow{\sim} T^*M$ .

## Examples

- $\mathbb{R}^{2n}$  admits a natural symplectic form  $\omega = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$ .
- Any Kähler manifold  $X$  is symplectic with respect to the Kähler form.
- Any projective manifold is symplectic with respect to the Fubini–Study metric.

Note that in the latter two examples, despite  $X$  being a complex manifold, the Kähler form  $\omega \sim dz_1 \wedge d\bar{z}_1 + \cdots + dz_n \wedge d\bar{z}_n$  is not holomorphic.

## Definition

A *holomorphic symplectic manifold* is a pair  $(X, \sigma)$ , where  $X$  is a Kähler manifold equipped with a closed non-degenerate 2-form  $\sigma \in H^0(X, \Omega_X^2)$ .

# Holomorphic symplectic manifolds II

## Definition

A *holomorphic symplectic manifold* is a pair  $(X, \sigma)$ , where  $X$  is a Kähler manifold equipped with a closed non-degenerate 2-form  $\sigma \in H^0(X, \Omega_X^2)$ .

Again, this gives an isomorphism  $\iota_\sigma : \mathcal{T}_X \xrightarrow{\sim} \Omega_X^1$ .



# Holomorphic symplectic manifolds II

## Definition

A *holomorphic symplectic manifold* is a pair  $(X, \sigma)$ , where  $X$  is a Kähler manifold equipped with a closed non-degenerate 2-form  $\sigma \in H^0(X, \Omega_X^2)$ .

Again, this gives an isomorphism  $\iota_\sigma : \mathcal{T}_X \xrightarrow{\sim} \Omega_X^1$ .

## Examples

- $\mathbb{C}^{2n}$  admits a symplectic form  $\begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$ .

## Definition

A *holomorphic symplectic manifold* is a pair  $(X, \sigma)$ , where  $X$  is a Kähler manifold equipped with a closed non-degenerate 2-form  $\sigma \in H^0(X, \Omega_X^2)$ .

Again, this gives an isomorphism  $\iota_\sigma : \mathcal{T}_X \xrightarrow{\sim} \Omega_X^1$ .

## Examples

- $\mathbb{C}^{2n}$  admits a symplectic form  $\begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$ .
- Complex tori  $\mathbb{C}^{2n}/\Lambda$ ,  $\Lambda \simeq \mathbb{Z}^{4n}$ , are symplectic.

## Definition

A *holomorphic symplectic manifold* is a pair  $(X, \sigma)$ , where  $X$  is a Kähler manifold equipped with a closed non-degenerate 2-form  $\sigma \in H^0(X, \Omega_X^2)$ .

Again, this gives an isomorphism  $\iota_\sigma : \mathcal{T}_X \xrightarrow{\sim} \Omega_X^1$ .

## Examples

- $\mathbb{C}^{2n}$  admits a symplectic form  $\begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$ .
- Complex tori  $\mathbb{C}^{2n}/\Lambda$ ,  $\Lambda \simeq \mathbb{Z}^{4n}$ , are symplectic.
- K3 surfaces  $S$ ,  $\pi_1(S) = \{1\}$ ,  $\Omega_S^2 \simeq \mathcal{O}_S$ , are symplectic.

# Holomorphic symplectic manifolds II

## Definition

A *holomorphic symplectic manifold* is a pair  $(X, \sigma)$ , where  $X$  is a Kähler manifold equipped with a closed non-degenerate 2-form  $\sigma \in H^0(X, \Omega_X^2)$ .

Again, this gives an isomorphism  $\iota_\sigma : \mathcal{T}_X \xrightarrow{\sim} \Omega_X^1$ .

## Examples

- $\mathbb{C}^{2n}$  admits a symplectic form  $\begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$ .
- Complex tori  $\mathbb{C}^{2n}/\Lambda$ ,  $\Lambda \simeq \mathbb{Z}^{4n}$ , are symplectic.
- K3 surfaces  $S$ ,  $\pi_1(S) = \{1\}$ ,  $\Omega_S^2 \simeq \mathcal{O}_S$ , are symplectic.
- Any product of the spaces above is symplectic.

# The decomposition theorem

We can generalize the K3 surface as follows:

## Definition

We say a compact holomorphic symplectic manifold  $(X, \sigma)$  is *irreducible* if it is simply connected and  $\bigoplus_{p=0}^{\dim X} H^0(X, \Omega_X^{2p}) = \mathbb{C}[\sigma]$ .

# The decomposition theorem

We can generalize the K3 surface as follows:

## Definition

We say a compact holomorphic symplectic manifold  $(X, \sigma)$  is *irreducible* if it is simply connected and  $\bigoplus_{p=0}^{\dim X} H^0(X, \Omega_X^{2p}) = \mathbb{C}[\sigma]$ .

## Beauville–Bogomolov Decomposition Theorem

Let  $(X, \sigma)$  be a compact holomorphic symplectic manifold. Then there exists a finite cover

$$\pi : T \times \prod_i X_i \rightarrow X,$$

where  $T$  is a complex torus and the  $X_i$  are irreducible holomorphic symplectic manifolds.

# The decomposition theorem

We can generalize the K3 surface as follows:

## Definition

We say a compact holomorphic symplectic manifold  $(X, \sigma)$  is *irreducible* if it is simply connected and  $\bigoplus_{p=0}^{\dim X} H^0(X, \Omega_X^{2p}) = \mathbb{C}[\sigma]$ .

## Beauville–Bogomolov Decomposition Theorem

Let  $(X, \sigma)$  be a compact holomorphic symplectic manifold. Then there exists a finite cover

$$\pi : T \times \prod_i X_i \rightarrow X,$$

where  $T$  is a complex torus and the  $X_i$  are irreducible holomorphic symplectic manifolds.

Since complex tori are somewhat well understood, we will focus on understanding the irreducible holomorphic symplectic (IHS)  $X_i$ .

## Definition

Let  $(X, \sigma)$  be a holomorphic symplectic manifold of dimension  $2n$ , and let  $Y \subseteq X$  be a subvariety.

- 1  $Y$  is *isotropic* if  $\sigma|_Y = 0$ .
- 2  $Y$  is *Lagrangian* if  $Y$  is maximally isotropic, i.e.  $\sigma|_Y = 0$  and  $\dim Y = n$ .



## Definition

Let  $(X, \sigma)$  be a holomorphic symplectic manifold of dimension  $2n$ , and let  $Y \subseteq X$  be a subvariety.

- 1  $Y$  is *isotropic* if  $\sigma|_Y = 0$ .
- 2  $Y$  is *Lagrangian* if  $Y$  is maximally isotropic, i.e.  $\sigma|_Y = 0$  and  $\dim Y = n$ .

In general, we study symplectic manifolds by finding Lagrangian subvarieties.

## Definition

Let  $(X, \sigma)$  be a holomorphic symplectic manifold of dimension  $2n$ , and let  $Y \subseteq X$  be a subvariety.

- 1  $Y$  is *isotropic* if  $\sigma|_Y = 0$ .
- 2  $Y$  is *Lagrangian* if  $Y$  is maximally isotropic, i.e.  $\sigma|_Y = 0$  and  $\dim Y = n$ .

In general, we study symplectic manifolds by finding Lagrangian subvarieties.

## Theorem (Greb–Lehn '14)

*Let  $f : X \rightarrow B$  be a proper surjective morphism,  $B$  a normal analytic space of dimension not equal to 0 or  $2n$ . Then the fibers of  $f$  are Lagrangian subvarieties and  $B$  is a projective variety of dimension  $n$ .*

## Definition

Let  $(X, \sigma)$  be a holomorphic symplectic manifold of dimension  $2n$ , and let  $Y \subseteq X$  be a subvariety.

- ①  $Y$  is *isotropic* if  $\sigma|_Y = 0$ .
- ②  $Y$  is *Lagrangian* if  $Y$  is maximally isotropic, i.e.  $\sigma|_Y = 0$  and  $\dim Y = n$ .

In general, we study symplectic manifolds by finding Lagrangian subvarieties.

## Theorem (Greb–Lehn '14)

*Let  $f : X \rightarrow B$  be a proper surjective morphism,  $B$  a normal analytic space of dimension not equal to 0 or  $2n$ . Then the fibers of  $f$  are Lagrangian subvarieties and  $B$  is a projective variety of dimension  $n$ .*

For this reason, we refer to any such morphism  $f : X \rightarrow B$  as a *Lagrangian fibration*. Understanding the geometry of Lagrangian fibrations will help us understand the geometry of IHS manifolds.

# The structure of the fibration

Let  $f^\circ : X^\circ \rightarrow B^\circ$  be the smooth fibers of  $f : X \rightarrow B$ . We need to understand

- the base  $B$ ,
- the general fiber  $X_b$ ,
- the action of  $\pi_1(B^\circ, b)$  on  $X_b$ .

# The structure of the fibration

Let  $f^\circ : X^\circ \rightarrow B^\circ$  be the smooth fibers of  $f : X \rightarrow B$ . We need to understand

- the base  $B$ ,
- the general fiber  $X_b$ ,
- the action of  $\pi_1(B^\circ, b)$  on  $X_b$ .

## Theorem

- (Matsushita '99) *The smooth fibers  $X_b$  are Abelian varieties.*

# The structure of the fibration

Let  $f^\circ : X^\circ \rightarrow B^\circ$  be the smooth fibers of  $f : X \rightarrow B$ . We need to understand

- the base  $B$ ,
- the general fiber  $X_b$ ,
- the action of  $\pi_1(B^\circ, b)$  on  $X_b$ .

## Theorem

- (Matsushita '99) *The smooth fibers  $X_b$  are Abelian varieties.*
- (Hwang '08) *If  $B$  is smooth,  $B \simeq \mathbb{P}^n$ .*

# The structure of the fibration

Let  $f^\circ : X^\circ \rightarrow B^\circ$  be the smooth fibers of  $f : X \rightarrow B$ . We need to understand

- the base  $B$ ,
- the general fiber  $X_b$ ,
- the action of  $\pi_1(B^\circ, b)$  on  $X_b$ .

## Theorem

- (Matsushita '99) *The smooth fibers  $X_b$  are Abelian varieties.*
- (Hwang '08) *If  $B$  is smooth,  $B \simeq \mathbb{P}^n$ .*

By Hodge theory, an Abelian variety  $A$  can be recovered from its cohomology:

$$A \simeq H^{1,0}(A)/(H^1(A, \mathbb{Z}))^\vee.$$

We will study the monodromy representation

$$\rho : \pi_1(B^\circ, b) \rightarrow \text{Aut}(H^1(X_b, \mathbb{Z})).$$

# Algebraic monodromy groups

## Definition

The *full algebraic monodromy group*  $M$  of  $\rho$  is the Zariski closure of  $\rho(\pi_1(B^\circ, b)) \subseteq \text{Aut}(H^1(X_b, \mathbb{Q})) = \text{Sp}_{2n, \mathbb{Q}}$ . Its identity component  $M^\circ \subseteq M$  is the *(connected) algebraic monodromy group*.



# Algebraic monodromy groups

## Definition

The *full algebraic monodromy group*  $M$  of  $\rho$  is the Zariski closure of  $\rho(\pi_1(B^\circ, b)) \subseteq \text{Aut}(H^1(X_b, \mathbb{Q})) = \text{Sp}_{2n, \mathbb{Q}}$ . Its identity component  $M^\circ \subseteq M$  is the *(connected) algebraic monodromy group*.

When  $X$  is projective, we can classify the possible representations of the algebraic monodromy group.

## Theorem (V '26+)

*Let  $X$  be a  $2n$ -dimensional projective IHS manifold admitting a Lagrangian fibration  $f : X \rightarrow B$ . The algebraic monodromy group  $M^\circ$  associated to  $f$  is one of the following groups:*

# Algebraic monodromy groups

## Definition

The *full algebraic monodromy group*  $M$  of  $\rho$  is the Zariski closure of  $\rho(\pi_1(B^\circ, b)) \subseteq \text{Aut}(H^1(X_b, \mathbb{Q})) = \text{Sp}_{2n, \mathbb{Q}}$ . Its identity component  $M^\circ \subseteq M$  is the (*connected*) *algebraic monodromy group*.

When  $X$  is projective, we can classify the possible representations of the algebraic monodromy group.

## Theorem (V '26+)

Let  $X$  be a  $2n$ -dimensional projective IHS manifold admitting a Lagrangian fibration  $f : X \rightarrow B$ . The algebraic monodromy group  $M^\circ$  associated to  $f$  is one of the following groups:

- 1  $M^\circ = \text{Sp}_{2n, \mathbb{Q}}$ , acting on  $H^1(X_b, \mathbb{Q})$  via the standard representation;

# Algebraic monodromy groups

## Definition

The *full algebraic monodromy group*  $M$  of  $\rho$  is the Zariski closure of  $\rho(\pi_1(B^\circ, b)) \subseteq \text{Aut}(H^1(X_b, \mathbb{Q})) = \text{Sp}_{2n, \mathbb{Q}}$ . Its identity component  $M^\circ \subseteq M$  is the (*connected*) *algebraic monodromy group*.

When  $X$  is projective, we can classify the possible representations of the algebraic monodromy group.

## Theorem (V '26+)

Let  $X$  be a  $2n$ -dimensional projective IHS manifold admitting a Lagrangian fibration  $f : X \rightarrow B$ . The algebraic monodromy group  $M^\circ$  associated to  $f$  is one of the following groups:

- 1  $M^\circ = \text{Sp}_{2n, \mathbb{Q}}$ , acting on  $H^1(X_b, \mathbb{Q})$  via the standard representation;
- 2  $M^\circ = (\text{SL}_{2, \mathbb{Q}})^n$ , acting on  $H^1(X_b, \mathbb{Q})$  diagonally;

# Algebraic monodromy groups

## Definition

The *full algebraic monodromy group*  $M$  of  $\rho$  is the Zariski closure of  $\rho(\pi_1(B^\circ, b)) \subseteq \text{Aut}(H^1(X_b, \mathbb{Q})) = \text{Sp}_{2n, \mathbb{Q}}$ . Its identity component  $M^\circ \subseteq M$  is the (*connected*) *algebraic monodromy group*.

When  $X$  is projective, we can classify the possible representations of the algebraic monodromy group.

## Theorem (V '26+)

Let  $X$  be a  $2n$ -dimensional projective IHS manifold admitting a Lagrangian fibration  $f : X \rightarrow B$ . The algebraic monodromy group  $M^\circ$  associated to  $f$  is one of the following groups:

- 1  $M^\circ = \text{Sp}_{2n, \mathbb{Q}}$ , acting on  $H^1(X_b, \mathbb{Q})$  via the standard representation;
- 2  $M^\circ = (\text{SL}_{2, \mathbb{Q}})^n$ , acting on  $H^1(X_b, \mathbb{Q})$  diagonally;
- 3  $M^\circ = \{1\}$ , acting trivially on  $H^1(X_b, \mathbb{Q})$ .

# Geometric consequences

In each of the cases above, we extract information regarding the geometry of  $X$  and  $f : X \rightarrow B$ .

# Geometric consequences

In each of the cases above, we extract information regarding the geometry of  $X$  and  $f : X \rightarrow B$ .

## Theorem (V '26+)

- 1 *When  $M^\circ = \mathrm{Sp}_{2n, \mathbb{Q}}$ , the very general fiber of  $f$  is a simple Abelian variety with no extra endomorphisms.*

# Geometric consequences

In each of the cases above, we extract information regarding the geometry of  $X$  and  $f : X \rightarrow B$ .

## Theorem (V '26+)

- 1 When  $M^\circ = \mathrm{Sp}_{2n, \mathbb{Q}}$ , the very general fiber of  $f$  is a simple Abelian variety with no extra endomorphisms.
- 2 When  $M^\circ = (\mathrm{SL}_{2, \mathbb{Q}})^n$ , there exists an elliptically fibered K3 surface  $g : S \rightarrow \mathbb{P}^1$  such that  $X \rightarrow B$  is birational to  $S^n / \mathfrak{S}_n \rightarrow \mathbb{P}^n$ . In particular, the fibers  $X_b$  are isogenous to products of elliptic curves.

# Geometric consequences

In each of the cases above, we extract information regarding the geometry of  $X$  and  $f : X \rightarrow B$ .

## Theorem (V '26+)

- 1 When  $M^\circ = \mathrm{Sp}_{2n, \mathbb{Q}}$ , the very general fiber of  $f$  is a simple Abelian variety with no extra endomorphisms.
- 2 When  $M^\circ = (\mathrm{SL}_{2, \mathbb{Q}})^n$ , there exists an elliptically fibered K3 surface  $g : S \rightarrow \mathbb{P}^1$  such that  $X \rightarrow B$  is birational to  $S^n / \mathfrak{S}_n \rightarrow \mathbb{P}^n$ . In particular, the fibers  $X_b$  are isogenous to products of elliptic curves.

The case (3) is also somewhat well understood based on previous work:

## Theorem (Kim–Laza–Martin '25, V '26)

When  $M^\circ = \{1\}$ , there exists an elliptic curve  $E$  for which the smooth fibers  $X_b$  are isogenous to  $E^n$ .



# Proof of the main theorem.

*Proof idea.* We have the following isomorphism of bundles:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{T}_{X^\circ/B^\circ} & \longrightarrow & \mathcal{T}_{X^\circ} & \longrightarrow & f^*\mathcal{T}_{B^\circ} \longrightarrow 0 \\ & & \parallel & & \parallel & & \parallel \\ 0 & \longrightarrow & f^*\Omega_{B^\circ}^1 & \longrightarrow & \Omega_{X^\circ}^1 & \longrightarrow & \Omega_{X^\circ/B^\circ}^1 \longrightarrow 0 \end{array}$$

The fibers of  $f$  are given by quotienting  $f_*\Omega_{X^\circ/B^\circ,b}^1$  by a full rank sublattice. In particular, the structure of the fibers of  $f$  comes from  $\mathcal{T}_B$ .

# Proof of the main theorem.

*Proof idea.* We have the following isomorphism of bundles:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{T}_{X^\circ/B^\circ} & \longrightarrow & \mathcal{T}_{X^\circ} & \longrightarrow & f^*\mathcal{T}_{B^\circ} \longrightarrow 0 \\ & & \parallel & & \parallel & & \parallel \\ 0 & \longrightarrow & f^*\Omega_{B^\circ}^1 & \longrightarrow & \Omega_{X^\circ}^1 & \longrightarrow & \Omega_{X^\circ/B^\circ}^1 \longrightarrow 0 \end{array}$$

The fibers of  $f$  are given by quotienting  $f_*\Omega_{X^\circ/B^\circ,b}^1$  by a full rank sublattice. In particular, the structure of the fibers of  $f$  comes from  $\mathcal{T}_B$ .

There exists a cover  $\pi : B' \rightarrow B$  for which  $\mathcal{T}_{B'} = \mathcal{F}_1 \oplus \cdots \oplus \mathcal{F}_k$  splits into transverse foliations  $\mathcal{F}_i$ , which are permuted by the covering group  $\mathfrak{S}_k$ .

# Proof of the main theorem.

*Proof idea.* We have the following isomorphism of bundles:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{T}_{X^\circ/B^\circ} & \longrightarrow & \mathcal{T}_{X^\circ} & \longrightarrow & f^*\mathcal{T}_{B^\circ} \longrightarrow 0 \\ & & \parallel & & \parallel & & \parallel \\ 0 & \longrightarrow & f^*\Omega_{B^\circ}^1 & \longrightarrow & \Omega_{X^\circ}^1 & \longrightarrow & \Omega_{X^\circ/B^\circ}^1 \longrightarrow 0 \end{array}$$

The fibers of  $f$  are given by quotienting  $f_*\Omega_{X^\circ/B^\circ,b}^1$  by a full rank sublattice. In particular, the structure of the fibers of  $f$  comes from  $\mathcal{T}_B$ .

There exists a cover  $\pi : B' \rightarrow B$  for which  $\mathcal{T}_{B'} = \mathcal{F}_1 \oplus \cdots \oplus \mathcal{F}_k$  splits into transverse foliations  $\mathcal{F}_i$ , which are permuted by the covering group  $\mathfrak{S}_k$ . The general leaves of  $\mathcal{F}_i$  are algebraic, yielding a generically finite rational map

$$p : B' \dashrightarrow Z^k.$$

# Proof of main theorem.

*Proof idea (continued).* Now quotient  $p : B' \dashrightarrow Z^k$  by  $\mathfrak{S}_k$ . We get a rational map  $X \xrightarrow{f} B \dashrightarrow Z^k/\mathfrak{S}_k$ . By a criterion of Saccà, we can construct another IHS manifold  $\tilde{X}$  birational to  $X$  with a morphism  $\varphi : \tilde{X} \rightarrow Z^k/\mathfrak{S}_k$  extending the composition.

# Proof of main theorem.

*Proof idea (continued).* Now quotient  $p : B' \dashrightarrow Z^k$  by  $\mathfrak{S}_k$ . We get a rational map  $X \xrightarrow{f} B \dashrightarrow Z^k/\mathfrak{S}_k$ . By a criterion of Saccà, we can construct another IHS manifold  $\tilde{X}$  birational to  $X$  with a morphism  $\varphi : \tilde{X} \rightarrow Z^k/\mathfrak{S}_k$  extending the composition.

By a theorem of Matsushita,  $Z^k/\mathfrak{S}_k$  has the same cohomology as  $\mathbb{P}^n$ . This forces

$$\mathrm{IH}^*(Z, \mathbb{Q}) = H^*(\mathbb{P}^1, \mathbb{Q}),$$

so that  $Z = \mathbb{P}^1$ , and  $k = n$ .

# Proof of main theorem.

*Proof idea (continued).* Now quotient  $p : B' \dashrightarrow Z^k$  by  $\mathfrak{S}_k$ . We get a rational map  $X \xrightarrow{f} B \dashrightarrow Z^k/\mathfrak{S}_k$ . By a criterion of Saccà, we can construct another IHS manifold  $\tilde{X}$  birational to  $X$  with a morphism  $\varphi : \tilde{X} \rightarrow Z^k/\mathfrak{S}_k$  extending the composition.

By a theorem of Matsushita,  $Z^k/\mathfrak{S}_k$  has the same cohomology as  $\mathbb{P}^n$ . This forces

$$\mathrm{IH}^*(Z, \mathbb{Q}) = H^*(\mathbb{P}^1, \mathbb{Q}),$$

so that  $Z = \mathbb{P}^1$ , and  $k = n$ . Now we may construct a K3 surface  $S$  and a family of elliptic curves  $g : S \rightarrow \mathbb{P}^1$  so that

$$\begin{array}{ccc} S^n/\mathfrak{S}_n & \overset{\sim}{\dashrightarrow} & \tilde{X} \\ & \searrow & \swarrow \varphi \\ & \mathbb{P}^n & \end{array}$$

# Questions